

$$y = x^2 + 5 \quad (0, 0)$$

$$D(x) = x^2 + f(x)^2$$

$$D'(x) = 2x + 2f(x)f'(x)$$

$$2x + 2(x^2 + 5) \cdot 2x = 0$$

$$4x^3 + 22x = 0$$

$$2x(2x^2 + 11) = 0$$

$$x = 0, x^2 = -\frac{11}{2} \text{ (no real root)}$$

$$(x, x^2 + 5) \rightarrow (0, 5)$$

$$d = \sqrt{(0-0)^2 + (5-0)^2}$$

$$= \sqrt{25}$$

$$= 5$$

the closest point of $(0, 0)$ to $y = x^2 + 5$

is $(0, 5)$ and distance is 5

$$y = x^2 + 5 \quad (-4, 0)$$

$$D(x) = (x+4)^2 + f(x)^2$$

$$D'(x) = 2(x+4) + 2f(x)f'(x)$$

$$2x + 8 + 2(x^2 + 5) \cdot 2x = 0$$

$$4x^3 + 22x + 8 = 0$$

$$2x^3 + 11x + 4 = 0$$

$$x \approx -0.35547$$

$$(x, x^2 + 5) \rightarrow (-0.35547, 5.12636)$$

$$d = \sqrt{(x+4)^2 + (y-0)^2}$$

$$= \sqrt{39.5622}$$

$$\approx 6.28985$$

$$y = x^2 + 5 \quad (2, 0)$$

$$D(x) = (x-2)^2 + f(x)^2$$

$$D'(x) = 2(x-2) + 2(x^2 + 5) \cdot 2x = 0$$

$$4x^3 + 22x - 4 = 0$$

$$2x^3 + 11x - 2 = 0$$

$$x \approx 0.18074$$

$$(x, x^2 + 5) \rightarrow (0.18074, 5.03267)$$

$$d \approx 5.35140$$

$$y = x^2 + 5 \quad (-8, 0)$$

$$D(x) = (x+8)^2 + f(x)^2$$

$$= 2(x+8) + 2(x^2 + 5) \cdot 2x = 0$$

$$4x^3 + 22x + 16 = 0$$

$$2x^3 + 11x + 8 = 0$$

$$x \approx -0.67208$$

$$(x, x^2 + 5) \rightarrow (-0.67208, 5.45169)$$

$$d = \sqrt{(x+8)^2 + (y-0)^2}$$

$$d \approx 9.13342$$

$$y = x^2 + 5 \quad (6, 0)$$

$$D(x) = (x-6)^2 + f(x)^2$$

$$4x^3 + 22x - 12 = 0$$

$$2x^3 + 11x - 6 = 0$$

$$x \approx 0.51990$$

$$(x, x^2 + 5) \rightarrow (0.51990, 5.27030)$$

$$d \approx 7.60313$$